Year/ Semester	**:**	**III-year B.Tech. I Semester**
Branch	**:**	**Open Electives (all branches except CSBS)**
Academic Year	**:**	**2022–'23**
Subject	**:**	**Professional Ethics and Human Values**
Subject Code	**:**	**19OE1HS01**

Key Prepared by

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PART-A

a) Morals are always positive, and values may be positive or negative. A moral may be defined as something pertaining to a custom or usage, a principle or habit with respect to right or wrong conduct, or something concerned with the principles of right conduct and distinction between right and wrong. A value may be defined as a relative worth or importance, or something that is held to deserve, or an assigned/estimated worth, or a desirable quality. (Any one of the definitions listed above, for each, may be considered complete.)

b) Ethics of character insist that ethics are developed by developing one's own moral character first. Once an individual develops the right character, ethics naturally flow into the individual's actions and thoughts. Aristotle's Virtue Theory (Aristotelianism) is a prime example of Ethics of character, and some individuals who exhibited the ethics of character include people like Lord Rama, Dharmaraja/ Yudhishthira, Swami Vivekananda, or Dr. A.P.J. Abdul Kalam. (None of the examples are needed but students may express their understanding through an example though not a sound definition and even that may be considered a valid answer.)

c) Moral Responsibility (includes Obligations, Conscientiousness, Liability and Accountability, Praiseworthiness/ Blameworthiness), Causal Responsibility, Job Responsibility, and Legal Responsibility are considered Professional Responsibilities.

d) Organisational behaviour is a huge subject in itself. Thus, the answer may mention concepts such as collegiality, loyalty, work ethic, moral autonomy, moral leadership, moral character, moral responsibility and/or other professional responsibilities. Work ethic also includes ideas of diversity, mutual assistance and cooperation, collaboration and partnership, discipline, *etc.*

e) Risk assessment refers to estimating risk. Risk is defined as the possibility of something bad happening some time. Assessing risk is important to assess safety, since safety is defined as a state of affairs where risks are known and acceptable. Risk may sometimes be overestimated or underestimated or even ignored fully/ partially. Risk assessment must consider these possibilities.

f) Globalisation brings forth various aspects such as Ethics, Culture, Work Culture, Diversity, Consumer Behaviour, Consumer Buying Power, Equity Participation, Value-addition to local communities, Law, Politics, Safety and Security, Intergovernmental agreements and treaties, Economy, Market fluctuations, and Advertising aspects. Ethics must be considered in nearly all of these aspects when considering globalisation.

PART-B

1. Privacy protects access to individuals and is protected by a universal human right to privacy of individuals. Confidentiality protects access to information and is required in certain professional settings such as medical, legal, or social settings. While confidentiality requires privacy, it applies only when there is confidence in the information, and is provided in a setting that requires protection/ privacy.

Parents monitoring their children is really a privacy invasion, indeed. However, it is assumed to be done in the interest of the children and their safety and security. Parents must also be keen to notice their children's traits, to understand that they have matured enough to not have a continuous monitor. Such traits can include how children decide what is right or what is wrong, particularly when no one is watching. Children's demonstration of decision-making skills in a variety of situations, small and large, should be evidence of their maturity or lack thereof and must lead to putting a stop to the monitoring of online/ real-world activities. Breach of privacy can increase risk to individuals at personal and professional levels. When the children are conscious about being monitored, the parents need to reconsider if they must be monitoring their kids, since that indicates that the children are trying to understand the world around them through active observation and not mere passive adoption, and that can lead to better moral (character) development. Since maturity does not depend on biological age *per se*, monitoring may not stop after certain age but must stop after noticing certain traits among the children, definitely. It may also be important to mention that the legal age for adulthood is 18 and thus parents monitoring their children after this age is not legally supported either.

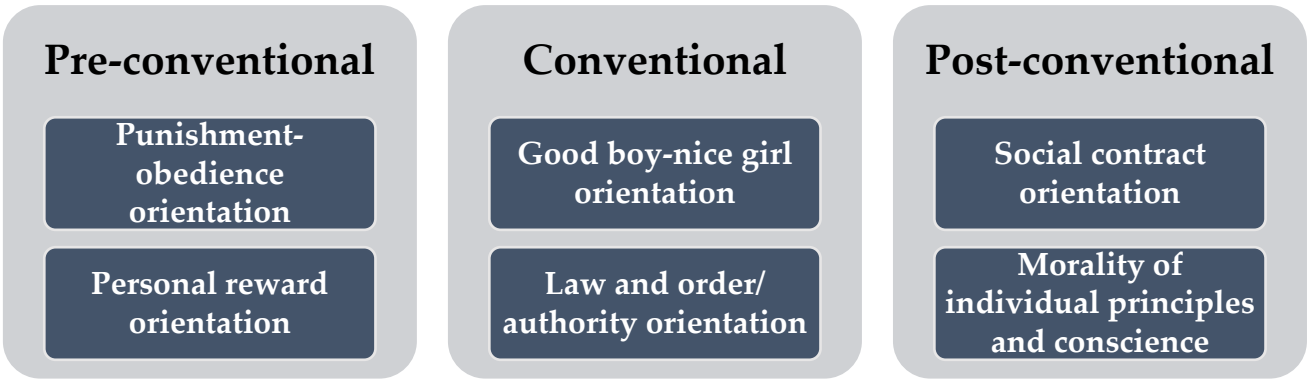
Note: Answers may include relevant application of theories such as the (Act) Utilitarianism or Kohlberg's/ Gilligan's moral development theory or concepts such as moral autonomy or moral character to elaborate and present their answer, but a mention of suitable theories or concepts is not a requirement for a good/ comprehensive answer.

2. a) A profession is defined as any job or occupation that requires advanced expertise, self-regulation, and a concerted effort towards public good. Engineering is aimed at providing for a better society in the future, through advancement of technology and facilitation. The role of a professional could be that of a social enabler, technology enabler, or a catalyst in getting the technology to the society. A professional may also be a saviour or guardian of public interest and an engineer shall act so, when public welfare is under threat. Engineering as a profession is thus justified through the roles an engineer plays, the traits an engineer has, and the responsibilities of an engineering, all of which align with that of a professional. Engineering as a profession discusses various roles of an engineer within a project too sometimes and involves ethical issues since the work of an engineer has direct and indirect impact on society and thus adds safety and risk assessment to an engineer's role.

   b) Moral judgement is a judgement based on morals while value judgement is a judgement based on a particular set of values/ value system. Value judgement is narrow in scope ("parochial") because it is based on a particular value system and is also a quick evaluation based on available information. Since not all information is available or authentic to make decisions, and since there can be biases in the value system followed/ believed/ applied, a value judgement can go wrong. Moral judgement is based on moral wisdom/ discernment/ discrimination. Moral deliberation has the capacity to go beyond a passive application of rules and requires appropriate sensitivity beyond rules to determine contextually about an

action or an actor (person). Moral judgement requires questioning the justification of the judgement, the basis for the judgement, and also questions if the individual making the moral judgement is worthy of making a moral judgement. Value judgement can be the same as a moral judgement if the individual practises morals for a long time and can make quick judgement based on available information based on morals and not just rules or values.

3. a) Kohlberg's Theory of Moral Development was the result of a study conducted on 1200 men. He proposed that moral development can happen in three levels, based on reasoning and motivation of individuals in response to moral questions, through social interaction. The three levels are: Pre-conventional, Conventional, and Post-conventional.



The right conduct is defined by direct benefit to oneself in pre-conventional level, by acceptance within the scope of law and authority in conventional level, and based on a comprehensive welfare for all in the post-conventional level. Pre-conventional level is typically seen in young children, conventional level is typical among adults, and only those adults or children with original thinking and critical reflection skills can mature enough to reach the post-conventional stage.

Kohlberg also mentions the 4+ level beyond the post-conventional level.

b) When resolving ethical dilemmas, consequences for all stakeholders including risks involved must be considered along with professional responsibilities and public-spiritedness, must be considered. In order to resolve moral dilemmas, it is important to identify moral factors/ reasons, collect relevant information/ facts/ data, rank the available moral options, generate alternate courses of action, discuss with colleagues/ experts, and decide based on a solution that harms the minimal and benefits the most. (The detailed procedure of resolving moral dilemmas is not a necessary part of the answer.)

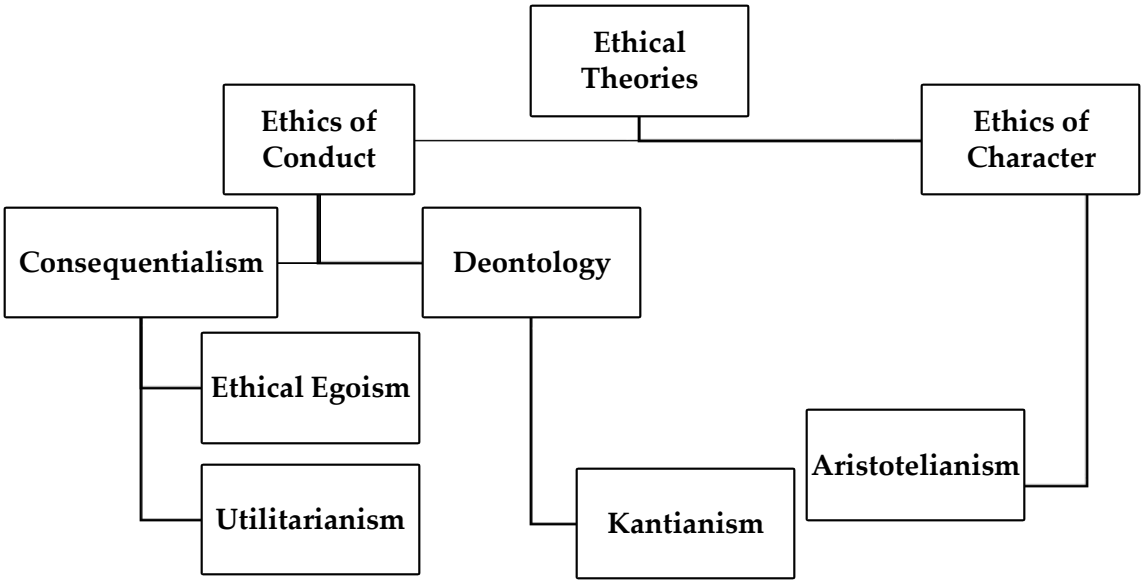
In addition to the above, different ethical inquiries lead us to resolving the ethical dilemmas by suitably applying the principles of ethical action and/or certain ethical theories. The three types of inquiries include:

- Normative: To identify the correct way of doing something ("How is something supposed to be done?")

- Conceptual: To clarify definitions, concepts, principles, and issues in engineering ethics ("What is what?")
- Factual: To describe current practices and facts of business pertaining to Value issues ("What exactly has been done or is to be done?")

Normative Inquiry refers to the description that describes what one ought to do under a specific circumstance. Conceptual Inquiry refers to the description of the meaning of concepts, principles and issues related to engineering ethics. Factual Inquiry or the descriptive inquiry helps provide the facts for understanding and finding solutions to value-based issues.

4. a) The following hierarchical chart presents how ethics are classified by character or by conduct.



Note: Students may choose to write a descriptive answer instead of a chart, indicating that the classification is based on when an action is right, or based on what an ethics theory's basis is. Even that may be considered the right answer if it is self-justifying.

b) Decision-making is needed at every micro- and macro-level in personal as well as professional lives. Without the right decision at the right time, the consequences can sometimes be trivial but can also be fatal or irreversible. Decision-making must suitably take into consideration ethics—various ethics theories help make the right decision in tough situations including moral/ ethical dilemmas.

Examples may vary from student and student and may be evaluated for inherent worth.

5. a) A social experiment is a scientific procedure undertaken to make a discovery, test a hypothesis, or demonstrate a known fact pertaining to the society. It is thus a research project conducted with human subjects in the real world. Since people are subjects of the experiment:
    - the rights of subjects and duties of experimenters are both important to be considered from ethics point of view,

- risk assessment and costs vs. benefits are both important,
- legal obligations of experimenters may limit innovation, and
- codes of ethics put primary responsibilities on experimenters.

Engineering may be seen as a social experimentation since all outcomes and consequences, benefits and risks, etc. are not always known. Aspects of engineering which are considered social experimentation involves design, production, installation, operation, as well as maintenance. Engineer must act as a Guardian of Public Interest and shares responsibility with the organization's management as well as the general public.

Note: Students may elaborate about an engineer's roles and responsibilities in the context of engineering being social experimentation. The answer may thus be evaluated in the spirit of its intrinsic worth and not based on specific content.

b) Intellectual Property covers artistic works such as music and literature, discoveries and inventions, words, phrases, symbols, and designs through rights and legal protection such as copyrights, patents, industrial design rights, and trademarks. Everything around us is IP, since it is born out of someone's intelligence and IP happens to be the prime motivator for Science, Technology, and Innovation (STI), since no innovation can motivate the innovators without its IP protected. A nation's industrialisation is measured by its number of patents. (Indian Patent Law is stricter than that in China, Korea, USA, or Japan, thus India has lesser patents but avoids disputes at WTO.) The objectives of IP law are to protect moral and economic rights of creators through financial incentive and maintaining economic growth as well as to respect morality. IP tends to be governed by economic goals when it should be viewed primarily as a social product. In order to serve human well-being, IP systems must respect and conform to human rights laws. When systems fail to do so, they risk infringing upon the human right to food and health, and to cultural participation and scientific benefits. World IP Organisation (WIPO) is established to encourage creative activity, to promote the protection of intellectual property throughout the world.

6. a) Non-disclosure Agreement (NDA) is a legally binding contract that establishes a confidential relationship and is aimed primarily to protect IP and work leading to IP rights. The parties that sign an NDA agree individually and/or mutually that any sensitive information they may obtain will not be made available to any others beyond the scope of the document. An NDA is thus usually signed before the parties discuss sensitive information. Some employers require some of their employees to sign an NDA, if the work leads to IP and thus sometimes includes patent-pending research work too.

   b) Professional traits include Collegiality, Loyalty/ Trustworthiness, Confidentiality, and Respect for Authority. Sense of Responsibility, Commitment, Conscientiousness, Self-improvement, Preparedness, and Leadership and Managerial Skills may also be considered additional professional traits.

Professional Rights is an elaborate list of the following, but not all may have to be explicitly listed by the examinees:

- Human Rights,
- Moral Rights,
- Intellectual Property Rights,
- Right to express professional judgement,
- Right to talk about one's work publicly within the bounds of confidentiality,
- Right to be associated with professional bodies and be involved in their activities,
- Right to protect clients and public from dangers of one's work,
- Right to professional recognition and due credit for one's services,
- Right of professional conscience to fulfil professional responsibilities, and
- Right to whistleblowing to preserve public safety and well-being.

Professional Responsibilities include:

- Moral Responsibility
  - Obligations
  - Conscientiousness
  - Liability and Accountability
  - Praiseworthiness/ Blameworthiness
- Causal Responsibility,
- Job Responsibility, and
- Legal Responsibility

Note: This question should have carried credits such that (a) may be for 2 marks and (b) may be for 6 marks. However, I failed to notice during the moderation that it was not mentioned thus in the question paper though I conveyed it orally. Evaluators are requested to be mindful of the same, in the sense that the first part of the answer may have too little to write, and the latter part may have much to write for 4 marks each. Due discretion is advised.

7. a) Organisational behaviour is a huge subject that encompasses concepts such as collegiality, loyalty, work ethic, moral autonomy, moral leadership, moral character, moral responsibility and/or other professional responsibilities. Work ethic also includes ideas of diversity, mutual assistance and cooperation, collaboration and partnership, discipline, *etc.* It is important for all employees to maintain the decorum of the organisation and comply by its policies, rules, and regulations in order to maintain a good behaviour within the organisation. Organisational behaviour thus encompasses interactions among people within the organisation and those with people outside the organisation in work-related setting. Employee rights and professional rights must be respected within the organisation and there must be enough scope for employees to perform their job responsibilities and for professionals to perform other responsibilities including moral responsibility, causal responsibility, and legal responsibility with due exercise of moral autonomy. Moral autonomy and respect for authority can co-exist through a robust and suitable ethical framework. In today's scenario, organisational behaviour includes interactions that are virtual and online due to the pandemic and requires the right etiquette in the changing scenario.

Note: Answers may mention workplace deviances and avoiding them, ethical climate, CEOs' mistakes, conflict resolution, work environment, complaint resolution, and more too. Evaluation must consider the intrinsic worth of the answer.

b) A clear conflict resolution procedure is needed for every organisation to ensure that conflicts do not affect corporate efficiency/ productivity and that a win-win situation prevails for organisation and its employees, balancing the authority with empathy. Conflicts can be because of schedules, priorities within projects/ departments, human resources, technical issues, administrative procedures, personalities, or costs vs. benefits. When conflicts are between people, a fair and empathetic approach with respect to individuals' autonomy is necessary. Conflicts of interest can be resolved by giving importance to organisational/ individual interests and not to positions, while considering all available options and choosing the best avenue that aligns with the organisational goal while ensuring fairness. Conflict resolution must keep in mind the short- and long-term consequences of the available options. Most ethical decisions have extended consequences, multiple alternatives, mixed outcomes, uncertain consequences, and personal implications. Managers must ensure that they consider all ethical and legal aspects involved and use suitable principles of ethical action based on the application of suitable ethical theories. (Students may use examples but examples are not mandatory.) Good managerial practices involve empathetic approach without prejudice towards anyone or anything.

8. Culpability is a measure of the degree to which an agent can be held morally or legally responsible for action and inaction or the degree of one's blameworthiness in the commission of a crime. It is characterised by direct involvement in the wrongdoing and not just for letting something happen due to failure to supervise. Thus, a culpable mistake is an intentional act by individual(s) without due justification, where the act as well as the consequences are known to the individual(s) and are controllable. Culpability may also arise due to acts that are thoughtless, insensitive, reckless, impulsive, short-sighted, and by assuming that what suits them will suit everyone instead of taking a more objective viewpoint, or even due to a misunderstanding of what is beneficial or harmful in the situation. Occupational crime is an illegal act by professional(s) using information and data that they gain as a part of their employment for self-interest or profit, also called "white-collar crimes", such as industrial espionage or manipulation of tenders or endangering lives. There may be minor deviances which are not criminal or culpable in nature too.
   Note: Suitable examples may be evaluated duly.
9. a) Welfare can refer to various aspects such as social status, social structure, culture, education, lifestyle, personal empowerment/situations, and income and wealth distribution. It is a professional responsibility (for engineers too) to be guardians of public interest and address ethical concerns pertaining to public welfare. It is important for professionals to inform the public about consequences of any activity and take the role of a saviour, guardian, or social servant in protecting public interest. Seeking informed consent from the affected public and/or being the voice of the impacted entities in the society are

part of addressing such ethical concerns. Whistleblowing is also a similar act of protecting public welfare.

Note: Examples can vary and may be evaluated for their intrinsic worth.

b) Bhopal gas tragedy refers to the leakage of methyl isocyanate (MIC) from the Union Carbide India Ltd. factory in Bhopal, Madhya Pradesh, India in 1984. The incident is considered the world's worst industrial disaster, which affected over 5.8 lakh people. The MIC-related safety systems in the plant were malfunctioning due to multiple issues and the gas leak was not contained even with multiple levels of safety measures. The abnormal pressure readings were thought to be instrumentation malfunction and a reported leak was not addressed immediately. Meanwhile, further leaks have affected the plant, eventually leaking over 40 tonnes of MIC within three hours from the gas leak that happened during midnight. A public siren was blown but shut down quickly before a longer one was blown. It was not until after two hours that the official information reached the nearest police station that there was a leak. However, it was reported that everything was under control. Even prior to this official acknowledgement, leaks were reported by the public but the Union Carbide personnel did not give correct information about which gas leaked and the local hospital could not be ready with any antidote for methyl isocyanate due to lack of information. Such acts of wilful negligence led to thousands of deaths by the morning and lakhs of people suffering from long-term effects such as cancers. Investigation was not conducted thoroughly, and the medical facilities were not equipped to handle the load either. It is thus important to keep business ethics, engineering ethics, and a timely risk assessment strategy in mind when developing policies in the future. All personnel, including law and order, industrial and medical personnel, must be trained duly to handle emergencies such as the tragedy and conscientiousness must be a part of professionalism.

Note: The case study and the subsequent question are presented elaborately above but the maximum credit of 4 marks for the answer may limit students' exposition of the answer. Due consideration may thus be given to students' overall understanding instead of specific information.

10. RMS Titanic was an avoidable accident. The disaster of Titanic happened over a century ago and is still relevant today in understanding the significance of ethics in the role of engineers. Most aspects of engineering safety and risk analysis were outcomes of the Titanic case. Learning from the past could have helped even the Titanic, if the White Star Line company that run the RMS Titanic learnt from its mistakes with the ship Olympic, which also met with an accident before Titanic was built. The failure of Titanic was on multiple aspects pertaining to ethics in engineering, management, and business!
    - **Failures in Engineering:**
        - Poorly cast wrought iron rivets caused steel plates on hull to come apart. They were inserted by hand, and the iron had more impurity ("slag"), which caused them to pop out upon collision.
        - The duration of testing was largely cut down!
        - The ship was built to withstand water filling in four of its compartments but six sunk in water!

- Ship could not be manoeuvred well through a complex interplay of two surface-water currents.
- Crucial ice warning from Mesaba not passed by senior radio operator.

- **Failures in Management:**
  - No pair of binoculars was provided in the crow's nest.
  - The duration of testing was largely cut down!
  - Getting people in lifeboats was not done efficiently.
  - Luxury was given more priority than safety!
  - High-speed steaming in a dangerous area caused the ice around the ship to melt, making it sink faster!
  - Production hastened to meet deadlines.
- **Failures of Business (Ethics):**
  - Lack of conscientiousness and integrity.
  - Regulatory requirements were falling short!
  - Lack of accountability and liability in the organisation's leadership.
  - Eye on publicity at the cost of making the ship safer!
  - Bad advertising created a lot of hype too, pressurizing the company to launch the ship faster!

11. **Note:** The answer to this question is purely subjective and must be evaluated for its intrinsic worth only without being judgemental towards any opinion. Students may choose to claim that being ethical in personal and professional lives is not necessary or practical. Students may claim that it is okay to tread off the ethical path once in a while and then come back on it "after the need is fulfilled" (and make up for lost ethics later) or even that they need not come back to the path (or that it is still wrong but unavoidable to veer off the path and that it does not matter even if one comes back). The answers (for both (a) and (b) parts) need to be evaluated with an open mind and look for any flaws in the justification that they believe in. An honest answer that supports being unethical and has a justification for it better than a dishonest answer that insists on being always ethical.
12. a) Students may pick any examples from the public domain about Dr. Abdul Kalam. Some of these may be "urban legends" (that may not have actually occurred in Dr. Kalam's life) that are viral on social media and the Web. The answers may be evaluated for their intrinsic worth. There may not be any examples for failures since they are not really failures in true sense but may be ethical dilemmas that Dr. Kalam faced or value judgements that Dr. Kalam made.

    b) Performance appraisals, Tight time-schedules, Fear of foreign competition, Management's attitude of considering profitability more important than consistent quality and customer retention are among the major threats of moral autonomy. Examples may be fairly aligned with these threats and may be from personal or professional lives.

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Scheme of Evaluation

- Students are expected to present insights based on the information provided and based on concepts learnt in the course. Thus, simplistic statements and general commentary may not be considered sufficient for any answer; students must articulate their learning through relevant keywords, concepts, theories, and logical bases as may be indicated by the corresponding Revised Bloom's Taxonomy level for each question.
- **The length of the answer does not define the correctness or rightness of the answer.** What is truly important is an understanding of relevant concepts and presentation of ideas and "connecting the dots". It is not sometimes surprising if a 12-mark question has an answer that is less than 1-page long, if the answer itself is self-explanatory and "feels" complete. It may be noted that answers by some students may be more elaborate or comprehensive than some others. Care may be taken to ensure that one student's answers do not influence another student's evaluation and the evaluation of one answer must not influence the evaluation of another answer by the same student.
- There are no fixed answers or fixed justifications/reasoning to any argument or case. Students may present their views on either side of an argument and may use one or more theories and concepts to justify their stance. It may even be possible that they present the arguments on both sides and/or present different ways of applying same concepts and theories. In any case, the evaluators may see the answers objectively and avoid all personal biases in judging the "rightness" of the answer. That is to say that ***answers may not seem* right *but may still be* correct**!
- Students may use one or more real or hypothetical examples (or counterarguments) to make their point clearer. Examples are only deemed necessary when the question specifies so but can be an added value even otherwise. Thus, an ill-defined but well-illustrated concept/ theory deserves the same or better credit as a well-defined concept that is not illustrated with suitable examples.
- Credit shall be suitably divided between (i) definitions of keywords used in the question/ answer, (ii) concept/ theory/ principle pertaining to morals/ ethics/ values, (iii) examples used to elaborate/ demonstrate the presented ideas, (iv) creative ways (including creativity in choosing the examples) of presenting the answer.
- This course is primarily taught with more emphasis on application of ethics and understanding of morals in the context of personal and professional lives, modulated to the level of comprehension of undergraduate students. The evaluation may thus consider not mere regurgitation of theory learnt from books, but the practical implications of the answers written by students. Students' answers may thus reflect their own understanding and may not partially or fully

match any specific textbook content or any other standard sources of content. Discretion and case-by-case evaluation is thus most suited for evaluation.

Disclosure/ Disclaimer:

This key is provided only to provide a generic/common approach to the questions posed in the examination. Thus, this key may not be treated as a comprehensive and all-inclusive guide for the answers. Owing to the generic nature of the subject and the questions, students may employ different approaches, presentation styles, or levels of elaboration. It is requested that the evaluator shall be accommodating towards all the diverse approaches that may seem valid at the outset, giving the benefit of doubt to the examinees/ students accordingly. Practical knowledge and applicability of various theories and concepts learnt in the subject may vary from student to student and can only be assessed when the evaluator puts one's own experience into understanding the presented answers.

*****End of Key*****